1.Hello World.

```
package JavaFoundation;
/**
 * HelloWorld
 */
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello World !");
    }
}
```

2.Variable.

```
package JavaFoundation;

public class Variable {
    public static void main(String[] args) {
        int var1 = 21;
        var1 = 5;
        var1 = var1 + 5;
        System.out.println(var1);
    }
}
```

3.Datatypes.

```
package JavaFoundation;

public class Datatypes {
    public static void main(String[] args) {
        int a = 24;
        String Name = "Varad";
        double b = 1.345678934567;
        float c = 123456.456f;
        boolean abc = true;
        byte f = 123;
        short g = 12345;
        char name = 'V';
        long sd = 12345612345234534561;

        System.err.println(a);
        System.err.println(b);
        System.err.println(c);
        System.err.println(abc);
        System.err.println(f);
        System.err.println(sd);
        System.err.println(name);
        System.err.println(Name);
        System.err.println(g);

    }
}
```

4.Increasing Pattern.

```
package PatternPrinting;

public class IncreasingPattern {
    public static void main(String[] args) {
        for (int i = 0; i <= 5; i++) {
            for (int j = 0; j <= i; j++) {
                System.out.print(" *");
            }
            System.out.println();
        }
    }
}
```

5.Right Angle Triangle.

```
package PatternPrinting;

public class RightAngleTriangle {
    public static void main(String[] args) {
        int n=5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <=n-i; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

6.Pyramid.

```
package PatternPrinting;

public class Pyramid {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= 2*i-1; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

7.Inverted Pyramid.

```
package PatternPrinting;

public class InvertedPyramid {
    public static void main(String[] args) {
        int n=5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <=i-1; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= 2*(n-i)+1; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

8.Hollow Square.

```
package PatternPrinting;

public class HolloSquare {
    public static void main(String[] args) {
        int n = 5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n; j++) {
                if (i == 1 || i == n || j == 1 || j == n) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

9.Arithmetic.

```
package BasicMathematics;

public class Arithmetic {

    public static void main(String[] args) {
        int a = 10;
        int b = 5;

        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
        System.out.println("Division: " + (a / b));
        System.out.println("Modulus: " + (a % b));
    }
}
```

10.GCD, HCF and LCM.

```
package BasicMathematics;

import java.util.Scanner;

public class GCD_HCFandLCM {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int num1 = a;
        int num2 = b;

        // Calculate GCD (HCF)
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }

        int gcd = a;
        int lcm = (num1 * num2) / gcd;

        System.out.println("GCD (HCF): " + gcd);
        System.out.println("LCM: " + lcm);

        sc.close();
    }
}
```

11.Array Basic.

```
package Arrays;

public class ArrayBasic {
    public static void main(String[] args) {
        int[] arr = { 10, 20, 30, 40, 50 };
        System.out.println(arr[0]);
        System.out.println(arr[arr.length - 1]);
        System.out.println(arr.length);
        arr[2] = 99;
        System.out.println(arr[2]);
    }
}
```

12.Traversal.

```
package Arrays;

public class Traversal {
    public static void main(String[] args) {
        int[] arr = { 10, 20, 30, 40, 50 };
        for (int i = 0; i < arr.length; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}
```

13.For-Each Loop.

```
package Arrays;

public class ForEach {
    public static void main(String[] args) {
        int[] arr = { 10, 20, 30, 40, 50 };
        for (int x : arr) {
            System.out.print(x + " ");
        }
    }
}
```

14.Input Output Array.

```
package Arrays;

import java.util.Scanner;
import java.util.Arrays;

public class InputOutputArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        System.out.println(Arrays.toString(arr));

        sc.close();
    }
}
```

15.Sum and Average.

```
package Arrays;

public class SumnAverage {
    public static void main(String[] args) {
        int[] arr = { 4, 7, 1, 9, 3 };
        int sum = 0;
        for (int x : arr) {
            sum = sum + x;
        }
        double avg = (double) sum / arr.length;
        System.out.println("Sum = " + sum);
        System.out.println("Average = " + avg);
    }
}
```

16.Min Max.

```
package Arrays;

public class MinMax {
    public static void main(String[] args) {
        int[] arr = { -5, -2, -9, -1 };
        int max = arr[0];
        int min = arr[0];
        for (int i = 1; i < arr.length; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
            if (arr[i] < min) {
                min = arr[i];
            }
        }
        System.out.println("Max = " + max);
        System.out.println("Min = " + min);
    }
}
```

17.Count Values That Appear.

```
package Arrays;

public class CountValuesAppears {
    public static void main(String[] args) {
        int[] arr = { 2, 5, 2, 8, 2 };
        int x = 2;
        int count = 0;
        for (int v : arr) {
            if (v == x) {
                count++;
            }
        }
        System.out.println(count);
    }
}
```

18.Linear Search.

```
package Arrays;
public class LinearSearch {
    public static void main(String[] args) {
        int[] arr = { 7, 3, 9, 4 };
        int key = 9;
        int index = -1;
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] == key) {
                index = i;
                break;
            }
        }
        System.out.println(index);
    }
}
```

19.Array References.

```
package Arrays;
import java.util.Arrays;
public class ArraysOfReference {
    public static void main(String[] args) {
        int[] a = { 1, 2, 3 };
        int[] b = a;
        b[0] = 99;
        System.out.println(a[0]);
        int[] c = Arrays.copyOf(a, a.length);
        c[1] = 50; System.out.println(a[1] + " " + c[1]);
    }
}
```

20.Two-Dimensional Arrays.

```
package Arrays;

public class TwoDarrays {
    public static void main(String args[]) {
        int[][] m = { { 1, 2, 3 }, { 4, 5, 6 }, { 7, 8, 9 } };
        for (int i = 0; i < m.length; i++) {
            int rowSum = 0;
            for (int j = 0; j < m[i].length; j++) {
                System.out.print(m[i][j] + " ");
                rowSum = rowSum + m[i][j];
            }
            System.out.println("| sum = " + rowSum);
        }
    }

}
```

21.Pos Neg Zero.

```
package Arrays;

public class PosNegZero {
    public static void main(String[] args) {
        int[] arr = { 5, -3, 0, 8, -1, 4 };
        int pos = 0, neg = 0, zero = 0;
        for (int x : arr) {
            if (x > 0) {
                pos++;
            } else if (x < 0) {
                neg++;
            } else {
                zero++;
            }
        }
        System.out.println("Positive: " + pos);
        System.out.println("Negative: " + neg);
        System.out.println("Zero: " + zero);
    }
}
```

22.Second Largest.

```
package Arrays;

public class SecondLargest {
    public static void main(String[] args) {
        int[] arr = { 12, 35, 1, 10, 34, 1 };
        int largest = Integer.MIN_VALUE;
        int second = Integer.MIN_VALUE;
        for (int x : arr) {
            if (x > largest) {
                second = largest;
                largest = x;
            } else if (x > second && x != largest) {
                second = x;
            }
        }
        System.out.println(second);
    }
}
```

23.Powers of 2 Less Than 100.

```
class PowersOf2LessThan100 {
    public static void main(String[] args) {
        int p = 1;

        while (p < 100) {
            System.out.print(p + " ");
            p = p * 2;
        }
    }
}
```

24.Print Even Numbers from 0 to 10.

```
class PrintEvenNumbers0To10 {
    public static void main(String[] args) {
        for (int i = 0; i <= 10; i += 2) {
            System.out.print(i + " ");
        }
    }
}
```

25.Break And Continue.

```
class BreakAndContinue {
    public static void main(String[] args) {
        for (int i = 1; i <= 10; i++) {
            if (i == 5) {
                continue;
            }

            if (i == 8) {
                break;
            }

            System.out.print(i + " ");
        }
    }
}
```

26.Demonstrate Do While.

```
class DemonstrateDoWhile {
    public static void main(String[] args) {
        int n = 10;

        do {
            System.out.println("Runs once, n = " + n);
        } while (n < 5);
    }
}
```

27.Multiplication Table Using Nested Loops.

```
class NestedLoopMultiplicationTable {
    public static void main(String[] args) {
        for (int i = 1; i <= 3; i++) {
            for (int j = 1; j <= 4; j++) {
                System.out.print(i * j + " ");
            }
            System.out.println();
        }
    }
}
```

28.Sum of Even Numbers from 1 to 10.

```
class SumOfEvenNumbers1To10 {
    public static void main(String[] args) {
        int sum = 0;

        for (int i = 2; i <= 10; i += 2) {
            sum += i;
        }

        System.out.println(sum);
    }
}
```

29.Count Doubling Steps.

```
class CountDoublingSteps {
    public static void main(String[] args) {
        int[] sizes = {1000, 1000000};

        for (int n : sizes) {
            int steps = 0;

            for (int i = 1; i <= n; i = i * 2) {
                steps++;
            }

            System.out.println("n = " + n + " -> " + steps + " steps");
        }
    }
}
```

30.Multiplication Table of 7.

```
class MultiplicationTableOf7 {
    public static void main(String[] args) {
        for (int i = 1; i <= 10; i++) {
            System.out.println("7 x " + i + " = " + 7 * i);
        }
    }
}
```

31.Four Shapes of Methods.

```
class FourMethodShapes {
    static void greet() {
        System.out.println("Hello!");
    }

    static void printStars(int n) {
        for (int i = 1; i <= n; i++) {
            System.out.print("*");
        }
        System.out.println();
    }

    static int add(int a, int b) {
        return a + b;
    }

    static boolean isEven(int n) {
        return n % 2 == 0;
    }

    public static void main(String[] args) {
        greet();
        printStars(4);
        System.out.println(add(3, 4));
        System.out.println(isEven(10));
    }
}
```

32.Square Using a Method.

```
class SquareMethod {
    static int square(int n) {
        return n * n;
    }

    public static void main(String[] args) {
        int result = square(5);
        System.out.println(result);
        System.out.println(square(12));
    }
}
```

33.Double All Array Elements.

```
import java.util.Arrays;

class DoubleAllArrayElements {
    static void doubleAll(int[] arr) {
        for (int i = 0; i < arr.length; i++) {
            arr[i] = arr[i] * 2;
        }
    }

    public static void main(String[] args) {
        int[] nums = {1, 2, 3};
        doubleAll(nums);
        System.out.println(Arrays.toString(nums));
    }
}
```

34.Find Index and Minimum/Maximum.

```
class FindIndexAndMinMax {
    static int findIndex(int[] arr, int key) {
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] == key) {
                return i;
            }
        }
        return -1;
    }

    static int[] minMax(int[] arr) {
        int min = arr[0], max = arr[0];
        for (int x : arr) {
            if (x < min) { min = x; }
            if (x > max) { max = x; }
        }
        return new int[]{min, max};
    }

    public static void main(String[] args) {
        int[] data = {7, 3, 9, 4};
        System.out.println(findIndex(data, 9));
        System.out.println(findIndex(data, 5));
        int[] r = minMax(data);
        System.out.println(r[0] + " " + r[1]);
    }
}
```

35.Pass-by-Value Swap Example.

```
class SwapWrongPassByValue {
    static void swapWrong(int a, int b) {
        int t = a;
        a = b;
        b = t;
    }

    public static void main(String[] args) {
        int x = 5, y = 10;
        swapWrong(x, y);
        System.out.println(x + " " + y);
    }
}
```

36.Call Stack with Squares.

```
class CallStackSquares {
    static int square(int n) {
        return n * n;
    }

    static int sumOfSquares(int a, int b) {
        return square(a) + square(b);
    }

    public static void main(String[] args) {
        System.out.println(sumOfSquares(3, 4));
    }
}
```

37.Method Overloading.

```
class MethodOverloading {
    static int add(int a, int b) {
        return a + b;
    }

    static double add(double a, double b) {
        return a + b;
    }

    static int add(int a, int b, int c) {
        return a + b + c;
    }

    public static void main(String[] args) {
        System.out.println(add(2, 3));
        System.out.println(add(2.5, 1.5));
        System.out.println(add(1, 2, 3));
    }
}
```

38.Scope Demo.

```
class ScopeDemo {
    static void demo() {
        int a = 10; // Lives until demo() ends.
        for (int i = 0; i < 3; i++) {
            int sq = i * i; // Lives for one pass only.
            System.out.println("i = " + i + ", square = " + sq);
        }
        System.out.println("a = " + a);
        // i and sq are out of scope here.
    }

    public static void main(String[] args) {
        demo();
    }
}
```

39.Maximum of Two Numbers Using a Method.

```
class MaxOfTwoMethod {
    static int max(int a, int b) {
        if (a > b) {
            return a;
        }
        return b;
    }

    public static void main(String[] args) {
        System.out.println(max(7, 12));
    }
}
```

40.Swap Array Elements Using a Method.

```
import java.util.Arrays;

class SwapArrayElements {
    static void swap(int[] arr, int i, int j) {
        int temp = arr[i];
        arr[i] = arr[j];
        arr[j] = temp;
    }

    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4};
        swap(arr, 0, 3);
        System.out.println(Arrays.toString(arr));
    }
}
```

41.Braces in an If Statement.

```
class BracesInIf {
    public static void main(String[] args) {
        int n = 3;
        if (n > 5) {
            System.out.println("Big");
        }
        System.out.println("Done?");
    }
}
```

42.Grade Using an Else-If Ladder.

```
class GradeWithElseIf {
    public static void main(String[] args) {
        int marks = 72;
        char grade;
        if (marks >= 90) {
            grade = 'A';
        } else if (marks >= 75) {
            grade = 'B';
        } else if (marks >= 60) {
            grade = 'C';
        } else if (marks >= 40) {
            grade = 'D';
        } else {
            grade = 'F';
        }
        System.out.println("Grade " + grade);
    }
}
```

43.Even or Odd Using If-Else.

```
class IfElseEvenOdd {
    public static void main(String[] args) {
        int n = 7;
        if (n % 2 == 0) {
            System.out.println("Even");
        } else {
            System.out.println("Odd");
        }
    }
}
```

44.Relational Operators.

```
class RelationalOperators {
    public static void main(String[] args) {
        int a = 7, b = 3;
        System.out.println(a > b);
        System.out.println(a == b);
        System.out.println(a != b);
        System.out.println(a <= b);
    }
}
```

45.Logical Operators.

```
class LogicalOperators {
    public static void main(String[] args) {
        int age = 25;
        char ch = 'a';
        int x = 3;
        System.out.println(age >= 18 && age <= 60);
        System.out.println(ch == 'a' || ch == 'e');
        System.out.println(!(x > 5));
    }
}
```

46.Character and Digit Check.

```
class CharacterAndDigitCheck {
    public static void main(String[] args) {
        char c = 'g';
        if (c >= 'a' && c <= 'z') {
            System.out.println("lowercase");
        }
        if (Character.isDigit('7')) {
            System.out.println("digit");
        }
    }
}
```

47.Classify a Character.

```
class ClassifyCharacter {
    public static void main(String[] args) {
        char[] tests = {'e', 'K', '7', '#'};
        for (char c : tests) {
            char lower = Character.toLowerCase(c);
            if (lower == 'a' || lower == 'e' || lower == 'i'
                    || lower == 'o' || lower == 'u') {
                System.out.println(c + " -> vowel");
            } else if (lower >= 'a' && lower <= 'z') {
                System.out.println(c + " -> consonant");
            } else if (c >= '0' && c <= '9') {
                System.out.println(c + " -> digit");
            } else {
                System.out.println(c + " -> other");
            }
        }
    }
}
```

48.Compare Decimal Values.

```
class DecimalComparison {
    public static void main(String[] args) {
        System.out.println(0.1 + 0.2 == 0.3);
        double x = 0.1 + 0.2;
        double y = 0.3;
        System.out.println(Math.abs(x - y) < 1e-9);
    }
}
```

49.Leap Year Check.

```
class LeapYearCheck {
    public static void main(String[] args) {
        int[] years = {2024, 1900, 2000, 2023};
        for (int year : years) {
            boolean leap = (year % 4 == 0 && year % 100 != 0)
                    || year % 400 == 0;
            System.out.println(year + " -> " + leap);
        }
    }
}
```

50.Positive, Negative or Zero.

```
class PositiveNegativeOrZero {
    public static void main(String[] args) {
        int[] nums = {5, -8, 0};
        for (int n : nums) {
            if (n > 0) {
                System.out.println(n + " -> Positive");
            } else if (n < 0) {
                System.out.println(n + " -> Negative");
            } else {
                System.out.println(n + " -> Zero");
            }
        }
    }
}
```

51.Safe Array Access.

```
class SafeArrayGet {
    static int safeGet(int[] arr, int i) {
        if (i < 0 || i >= arr.length) {
            return -1;
        }
        return arr[i];
    }

    public static void main(String[] args) {
        int[] arr = {10, 20, 30};
        System.out.println(safeGet(arr, 2));
        System.out.println(safeGet(arr, 7));
    }
}
```

52.Short-Circuit Guard.

```
class ShortCircuitGuard {
    public static void main(String[] args) {
        int n = 0;
        if (n != 0 && 100 / n > 5) {
            System.out.println("big");
        } else {
            System.out.println("safe");
        }
    }
}
```

53.Compare Strings.

```
class StringEquality {
    public static void main(String[] args) {
        String s1 = "hi";
        String s2 = new String("hi");
        System.out.println(s1 == s2);
        System.out.println(s1.equals(s2));
    }
}
```

54.Switch Arrow Form.

```
class SwitchArrowForm {
    public static void main(String[] args) {
        int day = 3;
        String name = switch (day) {
            case 1 -> "Mon";
            case 2 -> "Tue";
            case 3 -> "Wed";
            default -> "Other";
        };
        System.out.println(name);
    }
}
```

55.Switch Day.

```
class SwitchDay {
    public static void main(String[] args) {
        int day = 3;
        switch (day) {
            case 1: System.out.println("Mon"); break;
            case 2: System.out.println("Tue"); break;
            case 3: System.out.println("Wed"); break;
            default: System.out.println("Other");
        }
    }
}
```

56.Switch Fall-Through.

```
class SwitchFallThrough {
    public static void main(String[] args) {
        int x = 2;
        switch (x) {
            case 1: System.out.println("one");
            case 2: System.out.println("two");
            case 3: System.out.println("three");
        }
    }
}
```

57.Ternary Operator.

```
class TernaryOperator {
    public static void main(String[] args) {
        int a = 8, b = 15;
        int max = (a > b) ? a : b;
        String type = (a % 2 == 0) ? "Even" : "Odd";
        System.out.println(max + " " + type);
    }
}
```

58.Triangle Classification.

```
class TriangleClassification {
    static String classify(int a, int b, int c) {
        if (a <= 0 || b <= 0 || c <= 0) {
            return "Invalid";
        }
        if ((long) a + b <= c || (long) a + c <= b || (long) b + c <= a) {
            return "Invalid";
        }
        if (a == b && b == c) {
            return "Equilateral";
        }
        if (a == b || b == c || a == c) {
            return "Isosceles";
        }
        return "Scalene";
    }

    public static void main(String[] args) {
        int[][] tests = {{3, 4, 5}, {5, 5, 5}, {5, 5, 8},
                {1, 2, 3}, {0, 4, 4}};
        for (int[] t : tests) {
            String result = classify(t[0], t[1], t[2]);
            System.out.println(t[0] + " " + t[1] + " " + t[2] + " = " + result);
        }
    }
}
```